

Veterinary Package Insert

TP-Amoclavu WSP Amoxicillin & Clavulanate Potassium 300/75 mg/g Powder

PRODUCT DESCRIPTION

A white to off-white colour powder, which contains 300 mg/g of amoxicillin trihydrate and 75 mg/g of clavulanate potassium in 100g sachet.

PHARMACODYNAMICS & PHARMACOKINETICS

A combination of a beta-lactam antibiotic with a beta-lactamase inhibitor helps to restore the potency of amoxicillin against strains producing beta-lactamases.

Amoxicillin is a bactericidal antibiotic, which acts by inhibiting the synthesis of bacterial cell walls during bacterial multiplication. It inhibits cross-linkage between the linear peptidoglycan polymer chains in the cell wall of Gram-positive bacteria. Broad spectrum penicillin antibiotic amoxicillin is active also against a limited range of Gram negative bacteria where the outer layer of bacteria cell-wall is made up of lipopolysaccharide and protein.

There are three principal mechanisms of resistance to beta-lactam antibiotics: production of beta-lactamases, alteration of the penicillin binding protein (PBP), diminished outer membrane permeation. One of the most important mechanism of resistance is the inactivation of penicillin antibiotic with β -lactam enzymes produced by certain bacteria. These enzymes cleave the penicillin beta-lactam ring and render the penicillin drug inactive. Clavulanic acid acts as inhibitor of bacterial beta-lactamases. It prevents destruction of the beta-lactam ring and penicillins by beta-lactamase enzymes. The reaction is irreversible and both the enzyme and the clavulanate are destroyed while the antibiotic activity is preserved.

The plasma pharmacokinetic properties of amoxicillin and clavulanic acid are relatively similar. Both compounds are stable in acid environment of the gastrointestinal tract. After oral administration, amoxicillin and clavulanic acid are readily absorbed. The absorption following oral administration does not appear to be inhibited by presence of food in the alimentary tract.

Both compounds penetrate well into diverse tissue fluids (pleural, synovial, peritoneal fluids) and inflammatory exudates but not through the blood-

brain barrier. Both compounds are largely eliminated by renal excretion.

INDICATION

Treatment of clinical infections:

- respiratory tract infections caused by *Actinobacillus pleuropneumoniae*, *Pasteurella multocida*, *Streptococcus suis*;
- gastrointestinal infections caused by *Clostridium perfringens*, *E. coli* and *Salmonella Typhimurium*; where the causative pathogens are beta-lactamase producing strains of bacteria susceptible to amoxicillin in combination with clavulanic acid and where clinical experience and/or susceptibility testing indicates the combination as the drug of choice.

RECOMMENDED DOSAGE AND ADMINISTRATION

Administration in drinking water.

Give 10mg of amoxicillin (in the form of trihydrate) and 2.5mg of clavulanic acid (in the form of potassium salt) per kg of body weight twice daily, i.e. 3.97g of TP-Amoclavu WSP Amoxicillin & Clavulanate Potassium 300/75 mg/g Powder 100 kg body weight twice daily. Treat for 5 days.

For the calculation of the amount to be administered every 12 hours the following formula can be used: Number of pigs x average body weight (kg) x dose rate (0.04 g of TP-Amoclavu WSP Amoxicillin & Clavulanate Potassium 300/75 mg/g Powder / kg body weight) twice daily. During the twice daily treatment periods, medicated drinking water should be the only water supply available. After all the medicated drinking water has been consumed, resume the supply of unmedicated water.

To ensure correct dosage body weight of animals should be determined as accurately as possible to avoid underdosing.

The intake of medicated drinking water depends on the clinical conditions of the animals as well as on the weather/temperature. In order to obtain the correct dosage the concentration of the product should be adjusted accordingly.

For bulk medication twice daily : Half the calculated total daily dosage of the product is scattered onto the surface of tepid water (approximately 20°C) and stirred until evenly dispersed. Add the required amount of water to achieve a concentration of 1.19g – 5.96g of TP-Amoclavu WSP Amoxicillin & Clavulanate Potassium 300/75 mg/g Powder per liter of drinking water and stir for 20 minutes to reach full solubility.

The administration of the medicated drinking water should be repeated every 12 hours.

Do not administer the product through a dosing pump (proportioner). Do not use a water acidifier simultaneously. After reconstitution the medicated drinking water has to be consumed within 24 hours. Do not use the product with water systems composed of metal.

CONTRAINDICATIONS

Do not use in case of hypersensitivity to penicillin or cephalosporin antibiotics or any of the excipients. The product should not be given to rabbits, guinea pigs, hamsters, gerbils or small herbivores.

WARNINGS & PRECAUTIONS.

The product should be given to pigs only.

i. Special precautions for use in animals

The uptake of medication by animals can be altered as a consequence of illness. In case of insufficient uptake of feed/ water animals should be treated parenterally. Use of the product should be based on susceptibility testing and should take into account official national and regional policies with respect to the use of broad spectrum antibiotics. Do not use in cases of bacteria sensitive to narrow spectrum penicillins or to amoxicillin as a single substance. Use of the product deviating from the instructions given in the package insert may increase the prevalence of bacteria resistant to amoxicillin and clavulanic acid, and may decrease the effectiveness of treatment with other β -lactam antibiotics, due to the potential for cross resistance. Due to the resistance rate detected in some countries in porcine isolates of *E. coli* against amoxicillin in combination with clavulanic acid the product should be used for the treatment of infections caused by *E. coli* only after susceptibility testing has been carried out. Administration of the product should not be used as a method to control nonclinical salmonella infections in pig herds. It is strictly recommended, that the product should not be used as a tool of salmonella control programmes. In the case of a history of MRSA on a farm it is inappropriate to use a combination of amoxicillin and clavulanic acid as there is a likelihood to co-select for MRSA. The use of the product should be combined with good management practices e.g. good hygiene, proper ventilation, no overstocking.

ii. Special precautions to be taken by the person administering the veterinary medicinal product to animals

Penicillins and cephalosporins may cause hypersensitivity (allergy) following injection, inhalation, ingestion or skin contact. Hypersensitivity to penicillins may lead to cross reactions to cephalosporins and vice versa. Allergic reactions to these substances may occasionally be serious.

Do not handle this product if you know you are sensitised or if you have been advised not to work with such preparations.

Handle this product with great care to avoid exposure, taking all recommended precautions.

If you develop symptoms following exposure such as a skin rash, you should seek medical advice and show the doctor this warning.

Swelling of the face, lips or eyes or difficulty with breathing are more serious symptoms and require urgent medical attention.

Avoid inhalation of dust. Wear either a disposable half-mask respirator conforming to European Standard EN149 or a non-disposable respirator to European Standard EN140 with a filter to EN143.

Wear gloves during preparation and administration of medicated water.

Wash any exposed skin after handling the product or medicated water.

Wash hands after use.

INTERACTIONS WITH OTHER MEDICATIONS

In general, penicillins may be inhibited by the antibiotics with bacteriostatic action such as macrolides, sulfonamides and tetracyclines. No specific data on interaction of the combination have been described in the veterinary literature available. Neomycin given orally inhibits the intestinal absorption of penicillin.

PREGNANCY AND LACTATION

Use only according to the benefit/risk assessment of the responsible veterinarian.

ADVERSE EFFECTS

It is known that adverse reactions including mild gastrointestinal signs (diarrhoea and vomiting) and allergic reactions (skin reactions, anaphylaxis) may occur after administration of penicillins.

OVERDOSE & TREATMENT

In case of severe hypersensitive reactions, the treatment should be discontinued and corticosteroids and adrenaline should be administered. Treatment should be symptomatic in other cases of adverse reactions.

WITHDRAWAL PERIOD

Meat and offal: 1 day

MAXIMUM RESIDUAL LIMIT (MRL)

All food producing species including swine

(Amoxicillin)

Muscle: 50 µg/kg

Liver: 50 µg/kg

Kidney: 50 µg/kg

Fat: 50 µg/kg

Swine (Clavulanic acid)

Muscle: 200 µg/kg

Fat: 200 µg/kg

Liver: 200 µg/kg

Kidney: 200 µg/kg

STORAGE CONDITION

Store in a cool, dry place (below 30°C). Protect from sunlight.

SHELF LIFE

2 years.

After 1st opening, reconstitution or dilution, use within 24 hours.

THYE PHARMA SDN. BHD.

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